

# AI Support + Sales Copilot

## Project Reference

Production-oriented support and sales chatbot with FastAPI, Next.js, document ingestion, grounded retrieval, and operator tooling.

### DATASET / SOURCE

Company handbook + uploaded PDFs

### CORE OUTPUT

Grounded answers + handoff flags

### STACK

FastAPI, Next.js, ChromaDB, Groq

### Business Problem

- Support and sales teams need fast answers that stay grounded in company documentation instead of improvising.
- Operators also need visibility into retrieval quality, document coverage, and when the assistant should escalate to a human.

### Approach

- Parse uploaded PDFs, split content into chunks, and index them into Chroma with a keyword fallback for constrained environments.
- Retrieve the most relevant chunks, rerank them, and answer with either a Groq-compatible LLM or a local extractive fallback.
- Expose health, retrieval, upload, list, and delete endpoints plus a demo-ready customer and admin UI.

### Key ML / AI Methods Used

- Retrieval-augmented generation with document chunking and context-window curation.
- Sentence-transformer embeddings with Chroma vector search plus a deterministic keyword fallback index.
- Extractive answer fallback and confidence-based human escalation when evidence is thin.
- API-first architecture with structured metadata for latency and retrieval counts.

### Results / Output

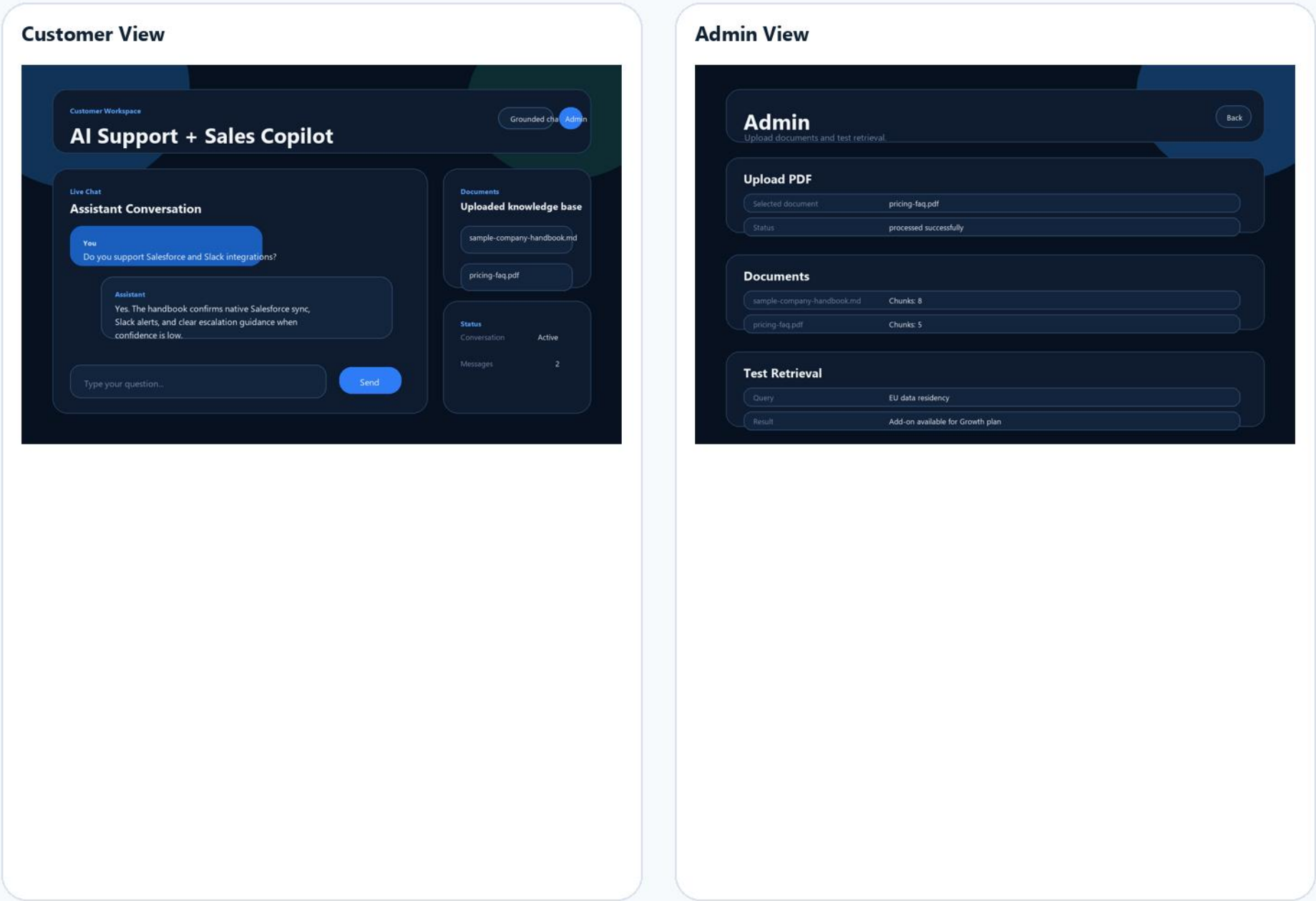
- Interactive customer-facing chat experience and an operator admin console for uploads, retrieval checks, and document lifecycle management.
- Backend API contract covering chat, health, document upload, retrieval, listing, and deletion.
- Robust local-demo mode: 10 backend tests passing, frontend production build passing, and resilience improvements for registry parsing, request flow, and fallback retrieval.
- GitHub: <https://github.com/JaiEnfer/ai-support-sales-copilot>

# Technology Stack

Backend: FastAPI, Pydantic Settings, PyPDF, LangChain text splitters  
AI / Retrieval: ChromaDB, sentence-transformers, Groq-compatible OpenAI client  
Frontend: Next.js 16, React 19, TypeScript, Tailwind CSS 4  
Quality: Pytest, linting, production build checks, sample chatbot flow validation

## Screenshots

Customer chat workspace and admin operations console



### Project Snapshot

- Project title: AI Support + Sales Copilot
- Business problem: grounded answers for support and sales teams using company knowledge sources.
- Dataset / source: sample company handbook, uploaded PDFs, and local fallback registry metadata.
- Approach: ingest, chunk, index, retrieve, rerank, answer, and escalate when confidence is low.
- Results / output: demo-ready UI, robust backend API, retrieval diagnostics, and local resilience when embeddings or external LLM access are unavailable.